

Hana Fujimoto

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Machine Learning Engineer (formerly civil engineering)

SKILLS

python, pytorch, onnx, opencv, cuda, numpy, ray, mlflow, polars

EXPERIENCE

Machine Learning Engineer *Sakura Vision* Osaka, Japan Apr 2023 – Present

- Ships defect-detection models to 40 factory edge boxes across seven client sites in Kansai and Chubu.
- Cut inference cost 60% by exporting to ONNX and quantising to INT8, with a calibration set per line.
- Owns the retraining loop: Ray for sweeps, MLflow for the registry, and a shadow-deploy gate before promotion.
- Rewrote the data layer on Polars; nightly aggregation went from 50 minutes to under four.
- Wrote the drift monitor that catches lens fouling before the false-positive rate reaches the operators.

Computer Vision Engineer *Rinku Robotics* Osaka, Japan Jan 2022 – Mar 2023

- Built the bin-picking pose estimator in PyTorch.

Machine Learning Intern *Naniwa Data Lab* Osaka, Japan Jul 2021 – Dec 2021

- Six months part-time alongside the bridge role, which is how the transition was funded.
- Built the baseline segmentation model later open-sourced as the bridge crack dataset.
- Presented weekly to a reading group of nine engineers on papers from CVPR and MVA.

Structural Engineer *Kansai Bridge & Structure* Kobe, Japan Apr 2014 – Dec 2021

- Load assessment for 60 prefectural road bridges under the 2016 revision of the Specifications for Highway Bridges.
- Led the seismic retrofit design for the Muko River crossing, a 340m continuous steel box girder.
- Ran finite-element models in Abaqus for fatigue life on orthotropic decks; published the calibration set.
- Supervised four junior engineers and the drawing office through two prefectural audits.
- Wrote the Python tooling that started the career change: a crack-width measurement pipeline over inspection photos that replaced two days of manual tracing per bridge.
- Standardised the inspection photo archive, 240k images, which later became the training set at Rinku.
- Represented the firm on the prefectural committee for bridge asset management, 2019-2021.

EDUCATION

MEng Civil Engineering, Kyoto University Apr 2012 – Mar 2014

Certificate in Applied Statistics, Osaka Institute of Technology Apr 2021 – Dec 2021

BEng Civil Engineering, Kyoto University Apr 2008 – Mar 2012

PROJECTS

Bridge crack segmentation `pytorch`, `opencv`

Open dataset and baseline; 1.1k GitHub stars.

CERTIFICATIONS

Professional Engineer, Civil (Japan) — JSCE, Mar 2018

Deep Learning Specialisation — Coursera, Sep 2021

JDLA Deep Learning for Engineers — JDLA, Jul 2022

PUBLICATIONS

Crack width estimation on orthotropic steel decks from inspection photography. *Journal of Bridge Engineering*, Jun 2021

Transfer learning from civil inspection imagery to industrial defect detection. *MVA*, Jul 2023

A calibration set for INT8 export of small segmentation models. *workshop paper*, Nov 2024